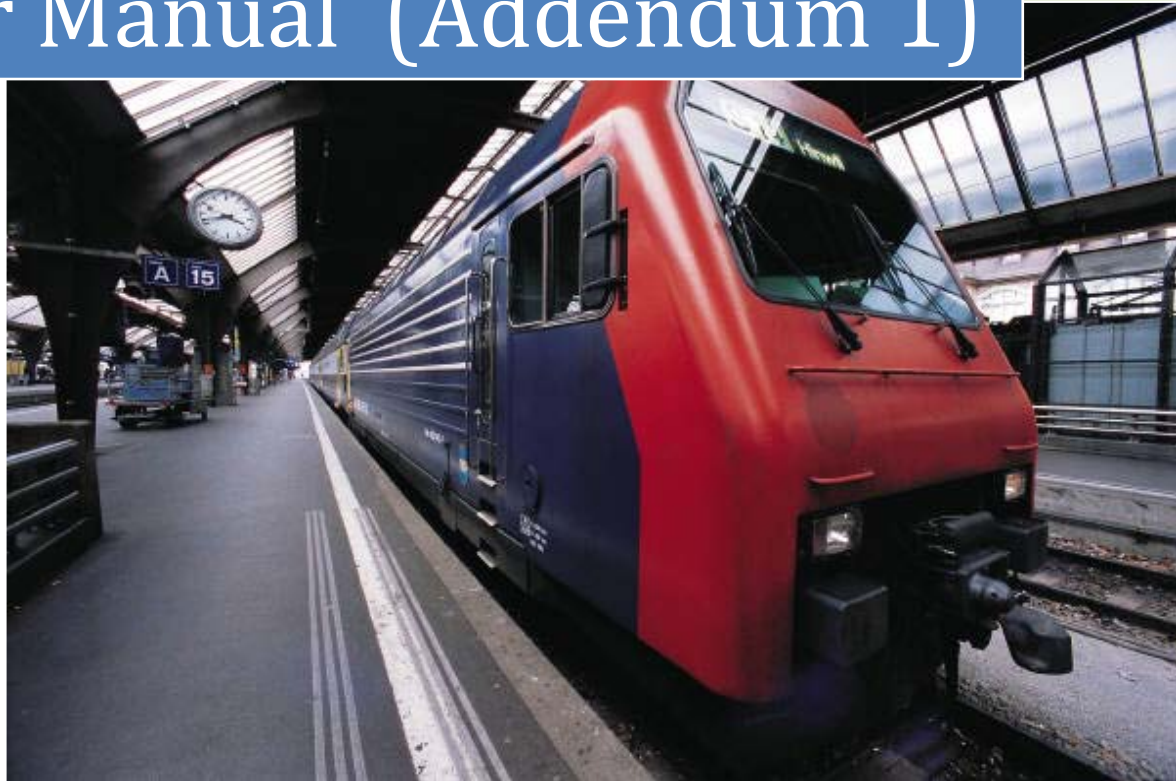


2020

cmOS Engineering LCP1 User Manual (Addendum 1)



cmOS Engineering

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Preliminary

Thank you for using a cmOS Engineering LCP1 Layout Command and Control (LCC) CAN node. Please ensure you read the sections on setting the jumpers and power requirements as a minimum before installing the LCP1 in your layout

This node is compliant with specifications of the OpenLCB project and offers a number of enhanced features.

Changes and Enhancements

The LCP1 power-up sequence has been changed to check for memory corruption and checking for user demand for firmware updating; 20 logic elements have been added to provide some additional simple signalling logic; and firmware update mode.

Logic Elements

Later versions of the LCP1 have 20 Logic Elements included.

The Logic Elements are configured in the same way as other Producer/Consumer events.

The logic element functions are:

- Timer - 0.5, 1.0, 2.0, 4.0 or 8.0 seconds;
- Start of Group;
- End of Group;
- AND;
- OR; and
- XOR.

NOT has not been provided as the Inactivate Event of the Output of Group is equivalent to the NOT function.

The timer functions are one shot per trigger. The trigger input is the Activate Event and the Inactivate Event is the time out of the Timer. To cause a timer to run continuously, make the Activate Event the same as the Inactivate Event. An initial Activate Event needs to be generated. Once a timer is activated in this way, it will run until the node is reset.

The combinational logic is a simple sequential evaluation, there are no precedence or parentheses evaluations.

The first input is the Start of Group and the eventual output is the End of group.

The Activate Event is the TRUE input to the selected logic function and the Inactivation Event is the False input.

Firmware Update

The LCP1 firmware maybe updated by the user. Before updating the node, ensure the configuration is backed up. This can be done using JMRI.

The update is accomplished by downloading a file from the cmOS Engineering Web site (www.cmoseng.com.au) and following the instructions given by the software. The node can be put into the update mode in several ways:

1. Pressing the reset button in the lower left corner. The first press will cause the node to go through its power-up sequence with the four LEDs cycling for about seconds. Once L2 is flashing, press the Reset Button again. This cycle must be repeated twice. Once the device is in the update mode, LEDs 3,4,5 will be on steady. If the Reset Button is pressed again while in this mode, the device will perform its power-up sequence again and revert to normal operation.
2. Use JMRI to initiate an update. Select ANY input file and select LOAD. The LEDs will agin all light and be steady.
3. If the node detects a fault during its power-up cycle, it will automatically enter the update mode.

Once the node is in the update mode, exit JMRI and run the file downloaded from the cmOS Engineering site.

Ensure only one node is in the update mode.

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In alphabetical order:

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OpenLCB group
Peter Fleury

References

cmOS Engineering: CMOSEng.com.au

Apply for node id range: <http://registry.openlcb.org/uniqueidranges>

General OLCB information: <http://openlcb.org/>

CAN documentation: https://en.wikipedia.org/wiki/CAN_bus

JMRI

Users: <https://groups.yahoo.com/neo/groups/jmriusers/conversations/messages>

Project: <http://jmri.sourceforge.net/>

NMRA Standards: <http://nmra.org/index-nmra-standards-and-recommended-practices>

Preliminary